

able for the purpose. Thus, $R_{lim} = 1.5k$ (which is resistor R5 in the circuit).

We can also manipulate the internal charge timer's expire values for IC1 by calculating the value of a resistor that will be connected from pin 14 (TMR) to ground. If resistor is not used at this pin, it selects the default value of 1800 seconds for pre-charge and 18,000 seconds for fast charge. We used the default timer values in this project. Here we complete the setup for our battery charging circuit.

The output (positive rail) from this section is taken from pins 10 and 11 (OUT) of IC1 and given to the next section via main switch S1. It is this switch which will be used to switch on the power supply.

Battery protection unit. This circuit deploys a low-power comparator LMP7300 (IC2) and a 3A PFET load switch TPS27082L (IC3) from Texas Instruments. It is basically an over-discharge or deep-discharge protection circuitry that continuously monitors the battery voltage levels. The battery should not be discharged beyond 2.7V. The over-discharge protection circuit not only shuts off the load connected to it but also alerts the user through LED indication that the battery is exhausted and needs recharging.

Voltage monitor. The LMP7300 is a combination comparator and reference with ideal specifications for precision threshold detection. The precision 2.048V reference comes with a 0.25% maximum error. It is mostly used for precision threshold detection, battery voltage monitoring and management system, and zero-crossing detectors.

The inverting input pin 2 of LMP7300 comparator is connected to reference voltage pin 7 (REF) while its non-inverting pin 1 is connected to a potential divider network comprising resistors R7 and R8. This potential divider is so designed that its output goes low when the input decreases to 2.8V (slightly higher than the 2.7V standard, to be on the safe side).

When the positive rail (which is the battery voltage) decreases to 2.8V, the voltage at pin 1 goes lower than

PARTS LIST

Semiconductors:	
IC1	- BQ24074 Li-ion battery charger and power-path manager
IC2	- LMP7300 precision comparator and reference
IC3	- TPS27082L PFET load switch
IC4	- PTN04050C adjustable boost converter
IC5	- UA78M33 positive-voltage regulator
IC6	- 7805, 5V regulator
LED1, LED2, LED7	- 5mm LED
LED3-LED6	- 3mm white LED (high intensity)
<i>Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):</i>	
R1, R2, R15	- 220-ohm
R3	- 10-kilo-ohm
R4	- 6-kilo-ohm
R5, R6	- 1.5-kilo-ohm
R7, R9	- 1-mega-ohm
R8	- 390-kilo-ohm
R10	- 4.7-kilo-ohm
R11	- 8.2-kilo-ohm
R12	- 470-kilo-ohm
R13	- 4.2-kilo-ohm
R14	- 68-ohm, 1W
Capacitors:	
C1	- 22 μ F, 25V electrolytic
C2	- 47 μ F, 25V electrolytic
C3	- 10nF ceramic disc
C4	- 100 μ F, 25V electrolytic
C5-C8	- 0.1 μ F ceramic
Miscellaneous:	
S1, S2	- On/off switch
CON1	- 3-pin connector
CON2	- 2-pin connector
CON3	- USB A-type connector
CON4	- 4-pin connector
	- DC Jack connector
	- 3.7V Li-ion battery
	- Shorting jumper
	- Heat sinks for IC5 and IC6

reference voltage at pin 2, which results in output of the comparator going low, thus stating under-voltage/over-discharge condition. The glowing of LED7 at output pin 4 of IC2 indicates the over-discharge or low-battery status.

Load switch. IC3 TPS27082L is a PFET load switch which switches on if its pin 5 (on/off) is given voltage greater than 1V. Pin 5 is directly connected to the output pin 4 of LMP7300. So whenever the battery output rail goes below 2.8V, the output of LMP7300 goes low, which switches off the load switch. Thus the load connected to this circuit is disconnected, hence no further discharging of the battery takes place.

Pull-up resistor R11 is provided to turn on the load switch IC3 normally because the output of the comparator

is open-drain. Resistor R12 and capacitor C3 in IC3 provide the slew rate of the load switch. The output from IC3 is taken from pins 2 and 3 (Vout) and then given to the boost converter section.

Boost converter. We used PTN04050C (IC4) boost converter module from Texas Instruments. It provides high-efficiency, step-up voltage conversion for loads of up to 12 watts. It is a DC-DC converter that steps up the low-voltage DC to a higher-voltage DC. It can take any voltage of 2.95V to 5.5V and step up to 9.2V. In fact, the boost converter can be configured to a maximum step-up voltage of 15V but, in our case, we step it up to 9.2V only, as that is sufficient for most portable power applications.

The positive rail taken from the previous section goes to pin 2 of IC4. Resistor R13 sets the output voltage of the converter to 9.2V, which is accessible at pin 4 (Vo) of IC4.

Regulator outputs. The 9.2V from IC4 is given to input pins of positive voltage regulators LMP7805 (IC6) and UA78M33 (IC5) to obtain 5V and 3.3V, respectively. Some capacitors are provided at the input and output of these regulators for stability purposes.

Thus, from a 3.7V Li-ion battery, we get regulated DC output voltages of 9.2V, 5V and 3.3V. These three voltages are routed to an external connector for the convenience of connecting to your projects.

The regulated (but not current-limited) 5V also goes to a female USB-A type connector so that you can power and charge most of your USB-powered products.

Torchlight. We have four white LEDs (LED3 through LED6) connected to 9.2V rail through resistor R14 and switch S2, serving the function of a torchlight/flashlight. A 68-ohm resistor (R14) is used as current limiter and switch S2 is used as on/off switch for the LEDs.

Construction and testing

An actual-size, single-side PCB for the mini rechargeable power supply